

Exponentials and Logarithms

A log answers one question: what power? Unknown in a power: take logs. Unknown inside logs: combine, then remove them.



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THE TOPIC ON ONE MAP

Exponentials and Logarithms

01 The laws

$\log ab, \log \frac{a}{b}$ $\log a^n$ in terms of k

02 Exponential equations

$a^x = b$ hidden quadratic unify the base

03 Log equations

one log each side number = log reject negatives

04 Graphs and models

$y = a^x$ asymptote $N = ka^t$

THE DEFINITION AND THE THREE LAWS

$$\log_a b = c \iff a^c = b \quad \log ab = \log a + \log b$$

$$\log \frac{a}{b} = \log a - \log b \quad \log a^n = n \log a$$

$$\log_a a = 1 \quad \log_a 1 = 0 \quad \log_a \sqrt{a} = \frac{1}{2} \quad \text{a number can be written as a log: } 3 = \log_2 8$$

01 The Laws of Logarithms: Simplify and Express in k

Break the expression into the pieces you are given: powers come out as multipliers, products become sums, quotients become differences. Replace each given log by its letter at the end.

WHEN YOU SEE → YOU DO

“given $\log_a b = k$, find $\log_a \left(\frac{\sqrt{a}}{b}\right)$ ”

$$\longrightarrow \frac{1}{2} \log_a a - \log_a b = \frac{1}{2} - k$$

“given $a = \log_2 3$, write $\log_2 9$ ”

$$\longrightarrow 9 = 3^2: 2a$$

“evaluate $\log_5 625$ ”

$$\longrightarrow \text{what power of 5 gives 625? } 4$$

“ $\log_{\sqrt{a}} a^6$ ”

$$\longrightarrow a^6 = (\sqrt{a})^{12}: 12$$

a Σ of logs

$$\longrightarrow \text{simplify one term, then sum (arithmetic)}$$

WORKED MODEL • OCTOBER 2022

$a = \log_2 3$. Write $\log_2 \left(\frac{\sqrt{3}}{16}\right)$ in terms of a .

1 $\log_2 \sqrt{3} - \log_2 16$

quotient law

2 $\frac{1}{2} \log_2 3 - 4$

power law; $16 = 2^4$

ANSWER $\frac{a}{2} - 4$

IDEA 01

Practise: The Laws of Logarithms: Simplify and Express in k

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Three laws

P2 · TOPIC 9 3 marks

Given $\log_a 2 = p$ and $\log_a 3 = q$, write in terms of p and q : (a) $\log_a 12$ (b) $\log_a \frac{9}{2}$ (c) $\log_a \sqrt{6a}$.

SELECTED PRACTICE

Q2**PRACTICE**

Practice · Easy

P2 · TOPIC 9 4 marks**1** Evaluate

(i) $\log_3 27$

(ii) $\log_5 625$

(iii) $\log_2 \frac{1}{4}$

(iv) $\log_a a$

(4 marks)

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 9**2 marks****12**

Use a calculator to find the value of

(i) $5 \log_3 7$

(ii) $2 \log_2 3 + 3 \log_3 2$

giving your answers to four significant figures.

(2 marks)**Q4****PRACTICE**

Practice · Very Hard

P2 · TOPIC 9**3 marks****4**

Without using a calculator, show that

$$\log_4 8 = \log_9 27.$$

(3 marks)

PAST PAPERS

Q5 **PAST PAPER** Pearson IAL WMA12 · October 2019 · Q7**P2 · TOPIC 9** **7 marks**

7. Given $\log_a b = k$, find, in simplest form in terms of k ,

(i) $\log_a \left(\frac{\sqrt{a}}{b} \right)$ (2)

(ii) $\frac{\log_a a^2 b}{\log_a b^3}$ (2)

(iii) $\sum_{n=1}^{50} (k + \log_a b^n)$ (3)

(Total 7 marks)

Q6**PAST PAPER**

Pearson IAL WMA12 · October 2022 · Q10

P2 · TOPIC 9**7 marks****10.** Given $a = \log_2 3$ (i) write, in simplest form, in terms of a ,

(a) $\log_2 9$

(b) $\log_2 \left(\frac{\sqrt{3}}{16} \right)$

(3)

(ii) Solve

$$3^x \times 2^{x+4} = 6$$

giving your answer, in simplest form, in terms of a .**(4)****(Total for Question 10 is 7 marks)**

02 Exponential Equations: Take Logs

Get the power on its own, then take logs of both sides: the power comes down in front. If the same base appears twice, as a^{2x} and a^x , it is a quadratic in disguise — substitute $y = a^x$.

WHEN YOU SEE → YOU DO

“solve $3^a = 70$ ”

$$\rightarrow a = \frac{\log 70}{\log 3}$$

“in the form $x = \log_7 a$ ”

$$\rightarrow 7^{x+2} = 3 \Rightarrow x + 2 = \log_7 3$$

$$5^{2x} - 8(5^x) + 12 = 0$$

$$\rightarrow y = 5^x: \text{quadratic, then } x = \log_5 y$$

“the curves $y = 3^{2x+1}$ and $y = 6 \times 3^x$ meet at P ”

$$\rightarrow \text{set equal, divide by } 3^x$$

$$8^{2x-5} = 20 \text{ in the form } a + b \log_2 5$$

$$\rightarrow \text{write } 8 = 2^3, \text{ then } \log_2 \text{ both sides}$$

$$y = a^x \text{ negative}$$

$$\rightarrow \text{no solution: } a^x > 0$$

Isolate the power

Take logs

Power comes down

Solve

WORKED MODEL · JUNE 2025

$3^{2x+1} = 6 \times 3^x$. Find x exactly.

$$1 \quad 3 \times (3^x)^2 = 6 \times 3^x$$

write 3^{2x+1} as $3 \cdot 3^{2x}$

$$2 \quad 3^x = 2$$

divide by $3 \cdot 3^x$

ANSWER $x = \log_3 2, \quad P = (\log_3 2, 12)$

IDEA 02

Practise: Exponential Equations: Take Logs

My questions first, then selected practice, then real past papers.

Q7**DR ESLAM**

Take logs

P2 · TOPIC 9**2 marks**

Solve $5^{2x-1} = 40$, giving x to 3 significant figures.

Q8**DR ESLAM**

A hidden quadratic

P2 · TOPIC 9**4 marks**

Solve $9^x - 10(3^x) + 21 = 0$.

Let $y = 3^x$: $9^x = y^2$.

SELECTED PRACTICE

Q9**PRACTICE** Practice · Easy

P2 · TOPIC 9 2 marks

4

Solve the equation

$$7^{2x-1} = 343.$$

(2 marks)

Q10**PRACTICE** Practice · Medium

P2 · TOPIC 9 3 marks

6

Solve $2^{2x} - 24(2^x) + 128 = 0$.

(3 marks)

Q11**PRACTICE**

Practice · Very Hard

P2 · TOPIC 9 4 marks**6**Solve $3^{2(x+1)} + 3 = 28(3^x)$.**(4 marks)**

PAST PAPERS

Q12 PAST PAPER Pearson IAL WMA12 · January 2021 · Q3

P2 · TOPIC 9 8 marks

3. (i) Solve

$$7^{x+2} = 3$$

giving your answer in the form $x = \log_7 a$ where a is a rational number in its simplest form.

(3)

(ii) Using the laws of logarithms, solve

$$1 + \log_2 y + \log_2 (y + 4) = \log_2 (5 - y)$$

(5)

(Total 8 marks)

Q13**PAST PAPER**

Pearson IAL WMA12 · October 2023 · Q5

P2 · TOPIC 9 6 marks

5. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve

$$3^a = 70$$

giving the answer to 3 decimal places.

(2)

(ii) Find the exact value of b such that

$$4 + 3 \log_3 b = \log_3 5b$$

(4)

(Total for Question 5 is 6 marks)

Q14 PAST PAPER Pearson IAL WMA12 · June 2025 · Q6

P2 · TOPIC 9 10 marks

- 6: **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

(i) Solve

$$2\log_2(4-x) = 3 + \log_2\left(\frac{x+11}{2}\right) \quad (5)$$

(ii) The curves C_1 and C_2 with equations

$$y = 3^{2x+1} \quad \text{and} \quad y = 6 \times 3^x$$

meet at the point P .

Find the exact coordinates of P , writing your answer in the form $(\log_3 a, b)$ where a and b are integers.

(5)

(Total for Question 6 is 10 marks)

Q15**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q7

P2 · TOPIC 9 9 marks

7.

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

- (i) Find the exact solution of the equation

$$8^{2x-5} = 20$$

giving the answer in the form $a + b \log_2 5$ where a and b are rational constants.

(4)

- (ii) Using the laws of logarithms, solve the equation

$$\log_3(13 + 2y) + 3 = 2\log_3(4 - y)$$

(5)

(Total for Question 7 is 9 marks)

03 Log Equations: One Log Each Side, Then Remove It

Use the laws to collect every log into a single log on each side. Write any plain number as a log of the same base. When it reads $\log_a(\dots) = \log_a(\dots)$, drop the logs. Finally, reject any answer that makes a log of a negative number.

WHEN YOU SEE → YOU DO

“ $\log_3(32 - 12x) = 2 \log_3(1 - x) + 3$ ”

→ $3 = \log_3 27$; $2 \log_3(1 - x) = \log_3(1 - x)^2$

a number alone on one side

→ $\log_a X = n \Rightarrow X = a^n$

different bases $\log_4$ and $\log_{16}$

→ $\log_{16} X = \frac{1}{2} \log_4 X$

“show that $2x^3 - 3x^2 - 30x + 56 = 0$ ”

→ remove the logs, expand, tidy

“hence solve”

→ check each root: every log argument > 0

WORKED MODEL • OCTOBER 2024

Solve $\log_4(12 - 2x) = 2 + 2 \log_4(x + 1)$.

1 $\log_4(12 - 2x) = \log_4\{16(x + 1)^2\}$

$2 = \log_4 16$

2 $16x^2 + 34x + 4 = 0 \Rightarrow (8x + 1)(x + 2) = 0$

remove the logs

ANSWER $x = -\frac{1}{8}$ ($x = -2$ gives $\log_4(-1)$)

× **TRAP** Keeping a root that makes a log argument negative or zero. Always check.

IDEA 03

Practise: Log Equations: One Log Each Side, Then Remove It

My questions first, then selected practice, then real past papers.

Q16 **DR ESLAM** Combine, then remove

P2 · TOPIC 9 4 marks

Solve $\log_2(x + 6) - \log_2 x = 3$.

One log on the left, then 2^3 on the right.

Q17 **DR ESLAM** Numbers become logs

P2 · TOPIC 9 4 marks

Solve $\log_3 x + \log_3(x - 8) = 2$.

SELECTED PRACTICE

Q18 **PRACTICE** Practice · Easy

P2 · TOPIC 9 2 marks

3 Solve the equation

$$\log_x 16 = 2$$

(2 marks)

Q19 **PRACTICE** Practice · Hard

P2 · TOPIC 9 3 marks

4 Solve the equation

$$2 \log_x (x + 2) = 3$$

giving your answer correct to 3 significant figures.

(3 marks)

Q20**PRACTICE**

Practice · Very Hard

P2 · TOPIC 9 7 marks**8**Solve the equation $\log_4(2 - x) = \log_{16}(13 - 4x)$.**(7 marks)**

PAST PAPERS

Q21 PAST PAPER Pearson IAL WMA12 · June 2021 · Q7

P2 · TOPIC 9 10 marks

7. (a) Given that

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25)$$

show that

$$2x^3 - 3x^2 - 30x + 56 = 0 \quad (4)$$

(b) Show that -4 is a root of this cubic equation. (2)

(c) Hence, using algebra and showing each step of your working, solve

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25) \quad (4)$$

(Total 10 marks)

Q22**PAST PAPER**

Pearson IAL WMA12 · January 2022 · Q4

P2 · TOPIC 9 5 marks

4. Using the laws of logarithms, solve

$$\log_3(32 - 12x) = 2 \log_3(1 - x) + 3 \quad (5)$$

(Total 5 marks)

Q23**PAST PAPER**

Pearson IAL WMA12 · June 2022 · Q4

P2 · TOPIC 9 6 marks

4. In this question you must show all stages of your working.

Give your answers in fully simplified surd form.

Given that a and b are positive constants, solve the simultaneous equations

$$\begin{aligned}a - b &= 8 \\ \log_4 a + \log_4 b &= 3\end{aligned}$$

(6)

(Total 6 marks)

Q24 **PAST PAPER** Pearson IAL WMA12 · January 2023 · Q4**P2 · TOPIC 9** **6 marks**

4. (i) Using the laws of logarithms, solve

$$\log_3(4x) + 2 = \log_3(5x + 7) \quad (3)$$

- (ii) Given that

$$\sum_{r=1}^2 \log_a(y^r) = \sum_{r=1}^2 (\log_a y)^r \quad y > 1, a > 1, y \neq a$$

find y in terms of a , giving your answer in simplest form.

(3)

(Total for Question 4 is 6 marks)

Q25 **PAST PAPER** Pearson IAL WMA12 · June 2023 · Q5**P2 · TOPIC 9** **5 marks**

5. Use the laws of logarithms to solve

$$\log_2(16x) + \log_2(x + 1) = 3 + \log_2(x + 6) \quad (5)$$

(Total for Question 5 is 5 marks)

Q26 **PAST PAPER** Pearson IAL WMA12 · January 2024 · Q6**P2 · TOPIC 9** **8 marks**

6. (a) Given that

$$2 \log_4 (x + 3) + \log_4 x = \log_4 (4x + 2) + \frac{1}{2}$$

show that

$$x^3 + 6x^2 + x - 4 = 0 \quad (4)$$

(b) Given also that -1 is a root of the equation

$$x^3 + 6x^2 + x - 4 = 0$$

(i) use algebra to find the other two roots of the equation. (3)

(ii) Hence solve

$$2 \log_4 (x + 3) + \log_4 x = \log_4 (4x + 2) + \frac{1}{2} \quad (1)$$

(Total for Question 6 is 8 marks)

Q27**PAST PAPER**

Pearson IAL WMA12 · June 2024 · Q3

P2 · TOPIC 9 6 marks

3. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) Using the laws of logarithms, solve

$$2\log_2(2-x) = 4 + \log_2(x+10) \quad (5)$$

- (ii) Find the value of

$$\log_{\sqrt{a}} a^6$$

where a is a positive constant greater than 1

(1)

(Total for Question 3 is 6 marks)

Q28**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q9

P2 · TOPIC 9 9 marks

9. Given that

$$3 \log_2(t + 4) - 2 \log_2(t - 2) = 7$$

(a) verify that $t = 4$ is a solution of the above equation,**(2)**

(b) show that

$$t^3 - 116t^2 + 560t - 448 = 0$$

(3)

(c) Hence, using algebra and showing your working, solve

$$3 \log_2(t + 4) - 2 \log_2(t - 2) = 7$$

giving each answer in simplest form.

*(Solutions based entirely on calculator technology are not acceptable.)***(4)****(Total for Question 9 is 9 marks)**

Q29**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q6

P2 · TOPIC 9 5 marks

6. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

Using the laws of logarithms, solve

$$\log_4(12 - 2x) = 2 + 2\log_4(x + 1)$$

(5)

(Total for Question 6 is 5 marks)

Q30**PAST PAPER**

Pearson IAL WMA12 · January 2025 · Q7

P2 · TOPIC 9 8 marks

7.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) The table below shows values of x and y , where $y = \log_{10}(x + 5)$, for x values between -1 and 4

x	-1	0	1	2	3	4
$y = \log_{10}(x + 5)$	$\log_{10} 4$	$\log_{10} 5$	$\log_{10} 6$	$\log_{10} 7$	$\log_{10} 8$	$\log_{10} 9$

Using the trapezium rule with all the y values in the given table, show that

$$\int_{-1}^4 \log_{10}(x + 5) \, dx \approx \log_{10} k$$

where k is an integer to be found.

(3)

- (ii) Find the value of a such that

$$2 \log_5(5 - a) - \log_5(a + 25) = 1$$

(5)

(Total for Question 7 is 8 marks)

Q31**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q9

P2 · TOPIC 9 10 marks**9.**

**In this question you must show detailed reasoning.
Solutions relying on calculator technology are not acceptable.**

(i) Solve

$$2\log_3(4x + 5) - \log_3(x + 3) = 2 \quad (5)$$

(ii) Given that $a > 0$, $b > 0$ and

$$\log_{10} a + \log_{10} b = \log_{10}(a + b)$$

(a) prove that $a = \frac{b}{b-1}$ (3)

(b) Hence write down the full restriction on the value of b , giving a reason for your answer. (2)

(Total for Question 9 is 10 marks)

04 Exponential Graphs and Models

$y = a^x$ ($a > 1$) rises through $(0, 1)$ with the asymptote $y = 0$; $y = a^{-x}$ is its reflection in the y -axis. A “ $+k$ ” moves the curve and its asymptote up k . In a model $N = ka^t$, k is the start value and a the growth factor.

WHEN YOU SEE → YOU DO

“sketch $y = 3 \times 4^x$ ”

→ through $(0, 3)$, asymptote $y = 0$

“sketch $y = \frac{1}{5^x} + 3$ ”

→ decreasing, through $(0, 4)$, asymptote $y = 3$

“ $y = 4^x - 6$ meets the x -axis”

→ $4^x = 6 \Rightarrow x = \log_4 6$

“ $w = \log_a(t + 5) - \log_a 4$, $w = 10$ when $t = 3$ ”

→ substitute, then solve for a

“find t in terms of w ”

→ undo the log: $a^w = \frac{t+5}{4}$

WORKED MODEL • JUNE 2026

Solve $\frac{1}{5^x} + 3 = 23$, to 3 significant figures.

1 $5^{-x} = 20$

isolate the power

2 $-x \log 5 = \log 20$

take logs

ANSWER $x = -\frac{\log 20}{\log 5} = -1.86$

IDEA 04

Practise: Exponential Graphs and Models

My questions first, then selected practice, then real past papers.

Q32 **DR ESLAM** Asymptote and intercept**P2 · TOPIC 9** **3 marks**

Sketch $y = 2^x - 8$. Give the intercepts and the asymptote.

SELECTED PRACTICE**Q33** **PRACTICE** Practice · Medium**P2 · TOPIC 9** **4 marks****4**

On the same axes, sketch the graphs of $y = 2^x$ and $y = 3^x$, labelling any points where the graphs cross the coordinate axes and writing down the equation of any asymptotes.

(4 marks)

Q34**PRACTICE**

Practice · Hard

P2 · TOPIC 9 5 marks**5 (a)**

- (a) On the same axes, sketch the graphs of $y = 4^x$ and $y = 5^x$.
Label any points of intersection with the coordinate axes.
Write down the equations of any asymptotes.

(4 marks)**(b)**

- (b) Write down an equation for the graph that is a reflection of $y = 4^x$ in the y -axis.

(1 mark)

Q35**PRACTICE**

Practice · Very Hard

P2 · TOPIC 9 5 marks**5 (a)**(a) On the same axes, sketch the graphs of $y = 0.3^x$ and $y = 0.5^x$.

Label any points of intersection with the coordinate axes.

Write down the equations of any asymptotes.

(4 marks)**(b)**(b) Write down an equation for the graph that is a reflection of $y = 0.5^x$ in the y -axis.**(1 mark)**

PAST PAPERS

Q36 PAST PAPER Pearson IAL WMA12 · January 2020 · Q9

P2 · TOPIC 9 7 marks

9. (a) Sketch the curve with equation

$$y = 3 \times 4^x$$

showing the coordinates of any points of intersection with the coordinate axes.

(2)

The curve with equation $y = 6^{1-x}$ meets the curve with equation $y = 3 \times 4^x$ at the point P .

- (b) Show that the x coordinate of P is $\frac{\log_{10} 2}{\log_{10} 24}$

(5)

(Total 7 marks)

Q37 PAST PAPER Pearson IAL WMA12 · June 2021 · Q2

P2 · TOPIC 9 8 marks

2.

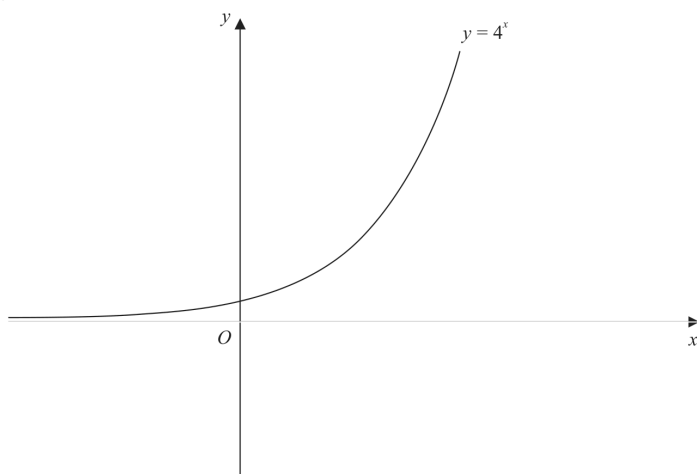
**Figure 1**

Figure 1 shows a sketch of the curve with equation $y = 4^x$

A copy of Figure 1, labelled Diagram 1, is shown on the next page.

(a) On Diagram 1, sketch the curve with equation

(i) $y = 2^x$

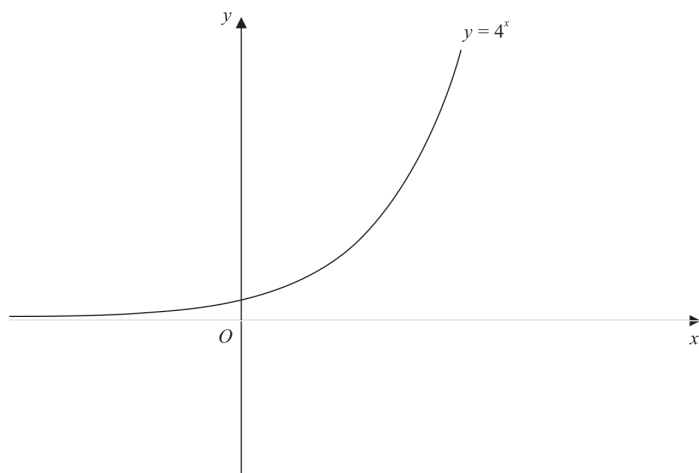
(ii) $y = 4^x - 6$

Label clearly the coordinates of any points of intersection with the coordinate axes.

(4)

The curve with equation $y = 2^x$ meets the curve with equation $y = 4^x - 6$ at the point P .

(b) Using algebra, find the exact coordinates of P .

(4)**2****Diagram 1****(Total 8 marks)**

Q38**PAST PAPER**

Pearson IAL WMA12 · October 2022 · Q4

P2 · TOPIC 9**8 marks**

4. The weight of a baby mammal is monitored over a 16-month period.

The weight of the mammal, w kg, is given by

$$w = \log_a(t + 5) - \log_a 4 \quad 2 \leq t \leq 18$$

where t is the age of the mammal in months and a is a constant.

Given that the weight of the mammal was 10 kg when $t = 3$

- (a) show that $a = 1.072$ correct to 3 decimal places.

(3)

Using $a = 1.072$

- (b) find an equation for t in terms of w

(3)

- (c) find the value of t when $w = 15$, giving your answer to 3 significant figures.

(2)

(Total for Question 4 is 8 marks)

Q39 **PAST PAPER** Pearson IAL WMA12 · June 2026 · Q6**P2 · TOPIC 9** **10 marks**

6. (i) (a) Sketch a graph of the curve with equation

$$y = \frac{1}{5^x} + 3 \quad x \in \mathbb{R}$$

Show on your sketch

- the coordinates of any points at which the curve crosses the axes
- the equation of any asymptotes

(3)

- (b) Using logarithms and making your method clear, solve the equation

$$\frac{1}{5^x} + 3 = 23$$

giving the answer to 3 significant figures.

(2)

- (ii) Given that $\log_4 x = a$, find in simplest form in terms of a

(a) $\log_4(64x)$

(2)

(b) $\log_4\left(\frac{2}{x^2}\right)$

(3)

(Total for Question 6 is 10 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------------|---|
| 01 | The laws | Powers out front; products add; quotients subtract. |
| 02 | Unknown in a power | Isolate, take logs; a^{2x} and $a^x \rightarrow$ quadratic. |
| 03 | Unknown in a log | One log each side, numbers as logs, remove, check. |
| 04 | Graphs, models | $(0, 1)$ and $y = 0$ moved by the transformation. |

BOTTOM LINE Power unknown $\rightarrow$ take logs · log unknown $\rightarrow$ remove the logs · always reject log of a negative.

Answers

Exponentials and Logarithms · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · THE LAWS OF LOGARITHMS: SIMPLIFY AND EXPRESS IN k

- Q1** Dr Eslam · Three laws (a) $2p + q$ (b) $2q - p$ (c) $\frac{1}{2}(p + q + 1)$
- Q2** Practice · Easy (i) 3 (ii) 4 (iii) -2 (iv) 1
- Q3** Practice · Easy (i) 8.856 (ii) 5.063
- Q4** Practice · Very Hard $4^{\frac{3}{2}} = 8$ and $9^{\frac{3}{2}} = 27$, so both equal $\frac{3}{2}$
- Q5** Pearson IAL WMA12 · October 2019 · Q7 (i) $\frac{1}{2} - k$. (ii) $\frac{k+2}{3k}$. (iii) $1325k$
- Q6** Pearson IAL WMA12 · October 2022 · Q10 (i)(a) $2a$. (i)(b) $\frac{a}{2} - 4$. (ii) $x = \frac{a-3}{a+1}$

IDEA 02 · EXPONENTIAL EQUATIONS: TAKE LOGS

- Q7** Dr Eslam · Take logs $2x - 1 = \frac{\log 40}{\log 5} = 2.292: x = 1.65$
- Q8** Dr Eslam · A hidden quadratic $y^2 - 10y + 21 = 0: y = 3 \text{ or } 7; x = 1 \text{ or } x =$
 $\log_3 7 \approx 1.77$
 $7^{2x-1} = 7^3 \Rightarrow x = 2$
- Q9** Practice · Easy
- Q10** Practice · Medium $y = 2^x: y^2 - 24y + 128 = 0, y = 8 \text{ or } 16: x = 3$
or 4
- Q11** Practice · Very Hard $y = 3^x: 9y^2 - 28y + 3 = 0, y = 3 \text{ or } \frac{1}{9}: x = 1 \text{ or}$
 $x = -2$
- Q12** Pearson IAL WMA12 · January 2021 · Q3 $(i) x = \log_7 \left(\frac{3}{49} \right). (ii) y = \frac{1}{2}$
- Q13** Pearson IAL WMA12 · October 2023 · Q5 $(i) a = 3.867. (ii) b = \frac{\sqrt{5}}{9}$
- Q14** Pearson IAL WMA12 · June 2025 · Q6 $(i) x = -2. (ii) P = (\log_3 2, 12)$
- Q15** Pearson IAL WMA12 · January 2026 · Q7 $(i) x = \frac{17}{6} + \frac{1}{6} \log_2 5. (ii) y = -5$

IDEA 03 · LOG EQUATIONS: ONE LOG EACH SIDE, THEN REMOVE IT

- Q16** Dr Eslam · Combine, then remove

$$\frac{x+6}{x} = 8 \Rightarrow x = \frac{6}{7}$$
- Q17** Dr Eslam · Numbers become logs

$$x(x-8) = 9 \Rightarrow x = 9 \text{ (} x = -1 \text{ rejected: log of a negative)}$$
- Q18** Practice · Easy

$$x^2 = 16 \Rightarrow x = 4 \text{ (a base must be positive)}$$
- Q19** Practice · Hard

$$(x+2)^2 = x^3 \Rightarrow x \approx 2.88$$
- Q20** Practice · Very Hard

$$\log_{16}(2-x)^2 = \log_{16}(13-4x) \Rightarrow x^2 = 9; x = -3$$

$$(x = 3 \text{ makes } 2-x < 0)$$
- Q21** Pearson IAL WMA12 · June 2021 · Q7
 (a) Proven. (b) -4 is a root of the cubic.
 (c) $x = 2, \frac{7}{2}$
- Q22** Pearson IAL WMA12 · January 2022 · Q4

$$x = -\frac{1}{9}$$
- Q23** Pearson IAL WMA12 · June 2022 · Q4

$$a = 4 + 4\sqrt{5}, \quad b = 4\sqrt{5} - 4$$
- Q24** Pearson IAL WMA12 · January 2023 · Q4
 (i) $x = \frac{7}{31}$. (ii) $y = a^2$
- Q25** Pearson IAL WMA12 · June 2023 · Q5

$$x = \frac{3}{2}$$
- Q26** Pearson IAL WMA12 · January 2024 · Q6
 (a) Proven. (b)(i) $x = \frac{-5 + \sqrt{41}}{2}$ and $x = \frac{-5 - \sqrt{41}}{2}$.
 (b)(ii) $x = \frac{-5 + \sqrt{41}}{2}$
- Q27** Pearson IAL WMA12 · June 2024 · Q3
 (i) $x = -6$. (ii) 12
- Q28** Pearson IAL WMA12 · June (R paper) 2024 · Q9
 (a) Verified. (b) Proven. (c) $t = 4$ or $t = 56 + 12\sqrt{21}$
- Q29** Pearson IAL WMA12 · October 2024 · Q6

$$x = -\frac{1}{8}$$
- Q30** Pearson IAL WMA12 · January 2025 · Q7
 (i) $k = 10080$. (ii) $a = -5$
- Q31** Pearson IAL WMA12 · October 2025 · Q2
 (i) $x = \frac{1}{16}$. (ii)(a) Proven. (ii)(b) $b > 1$

IDEA 04 · EXPONENTIAL GRAPHS AND MODELS

- Q32** Dr Eslam · Asymptote and intercept
y-intercept -7; x-intercept 3; asymptote $y = -8$
- Q33** Practice · Medium
both through $(0, 1)$, asymptote $y = 0$; 3^x above 2^x for $x > 0$, below for $x < 0$
- Q34** Practice · Hard
(a) both through $(0, 1)$, asymptote $y = 0$; 5^x steeper for $x > 0$ (b) $y = 4^{-x}$
- Q35** Practice · Very Hard
(a) both decreasing through $(0, 1)$, asymptote $y = 0$; 0.3^x below 0.5^x for $x > 0$ (b) $y = 0.5^{-x} = 2^x$
- Q36** Pearson IAL WMA12 · January 2020 · Q9
(a) Increasing exponential curve through $(0, 3)$, with no x-axis intersection. (b) $x = \frac{\log_{10} 2}{\log_{10} 24}$
- Q37** Pearson IAL WMA12 · June 2021 · Q2
(a)(i) Sketch $y = 2^x$ through $(0, 1)$ with no x-axis intercept. (a)(ii) Sketch $y = 4^x - 6$ through $(0, -5)$ and $(\log_4 6, 0)$. (b) $P = (\log_2 3, 3)$
- Q38** Pearson IAL WMA12 · October 2022 · Q4
(a) $a = 1.072$ to 3 d.p. (b) $t = 4(1.072)^w - 5$. (c) $t = 6.35$ months to 3 s.f
- Q39** Pearson IAL WMA12 · June 2026 · Q6
(i)(a) decreasing curve through $(0, 4)$, asymptote $y = 3$ (b) $5^{-x} = 20 \Rightarrow x = -1.86$ (ii)(a) $3 + a$ (b) $\frac{1}{2} - 2a$